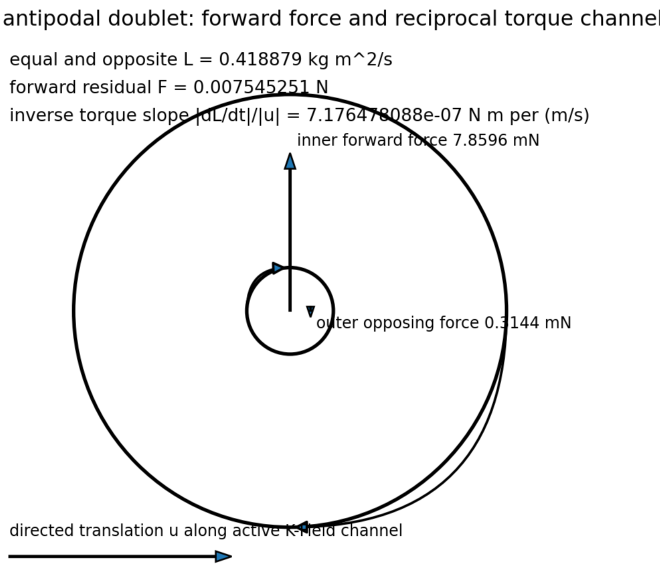
K-FIELD

K-Field Counter-Rotating System Inversion

Reciprocal conversion between directed translation and antipodal doublet rotation



Foundational reciprocal geometry: odd doublet rotation produces even linear force, and directed translation produces odd doublet torque through the transpose channel.

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This paper is a standalone foundational education and archival reference. It develops the inverse of the counter-rotating antipodal-doublet response from the supplied Cell3 geometry, the adopted axisymmetric K-Field rotor law, the reaction-constrained rotor calculation, and a declared minimal lossless reciprocal completion. Source-established statements, exact algebraic consequences, reciprocal-model deductions, and experimental predictions are kept distinct throughout.

Central result
The counter-rotating doublet is a reversible linear-rotational transducer on its active subspace. The source law maps odd doublet angular momentum into an even transverse force. The minimal local lossless reciprocal completion maps directed transverse translation into equal-and-opposite doublet torque.

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Abstract

A reaction-balanced pair of unequal concentric rotors can carry exactly equal and opposite angular momentum while producing a nonzero lateral K-Field force. This occurs because mechanical angular momentum is weighted by beta m r^2 omega, whereas the adopted lateral response is weighted by m omega. The forward map is therefore controlled by reciprocal specific areal inertia, 1/(beta r^2), rather than by angular momentum alone.

The forward relation is algebraically invertible wherever the geometric contrast and transverse orientation are nonzero. Algebraic invertibility does not by itself define the time evolution of the reverse process. The source ledgers specify force from rotation but do not specify torque from translation. A minimal reciprocal completion is therefore derived by requiring locality, linearity, the same geometric coefficient in both directions, reversal symmetry, and exact power conservation. The resulting equations are M udot=G L and A Ldot=-G u, where u is velocity along the active transverse direction, L is the odd doublet angular momentum, A=1/I_i+1/I_o is the rotational energy metric, and G is the source forward coefficient F/L.

Under motion control, the accumulated doublet angular momentum is proportional to directed displacement: Delta L=-(G/A) Delta x. Under force control, translation and counterrotation form a harmonic exchange system with angular frequency |G|/sqrt(MA). The worked source geometry gives G=0.01801296 per metre-second, reciprocal torque slope 7.1764781e-7 N m per m/s, and a free exchange period of approximately 16.10 hours for the two-rotor mass alone. These are device-specific values, not astronomical harmonics.

The centered Cell3 registry supplies a 13-dimensional odd inversion sector. At fixed field orientation, the spatial cross-product map has rank at most two, so no more than two odd Cell3 combinations can be bright reciprocal exchange modes at one orientation and at least eleven are dark. Rotation of the apparatus or commutation among internal channels changes the coupling map and can rotate dark modes into the active subspace.

Contents

1. Scope, objects, and evidentiary layers

The objective is to determine whether the established counter-rotating lateral-force process admits a physically meaningful inverse: whether directed acceleration and traversal through the K-Field can generate rotation of an antipodal doublet. The analysis begins with the exact source force law and reaction-constrained rotor state, establishes the algebraic inverse, then derives the simplest energy-conserving reciprocal dynamics consistent with that forward law.

Four statement classes are used. Source-established results are values or equations contained in the supplied machine-readable ledgers. Exact consequences follow algebraically from those source equations. Reciprocal-completion results follow from an explicitly stated dynamical closure. Experimental predictions follow from solving that closure with the actual source geometry. This separation is essential because the source archive does not independently state the inverse torque law.

Layer	Meaning
Source-established	Directly supplied geometry, field vector, rotor parameters, forward force law, and worked state.
Exact algebraic consequence	Null spaces, active transverse inverse, parity structure, and dimensional relations.
Reciprocal completion	Minimal local lossless translation-to-doublet torque law.
Experimental prediction	Time histories and scaling laws computed from the reciprocal equations.

2. Authoritative Cell3 geometry

Cell3 is represented by three nonorthogonal vectors a, b, and c. Their lengths, pairwise angles, face areas, and cell volume define the metric geometry. A canonical Cartesian realization preserves the supplied Gram matrix and is used for dimensionally correct figures.

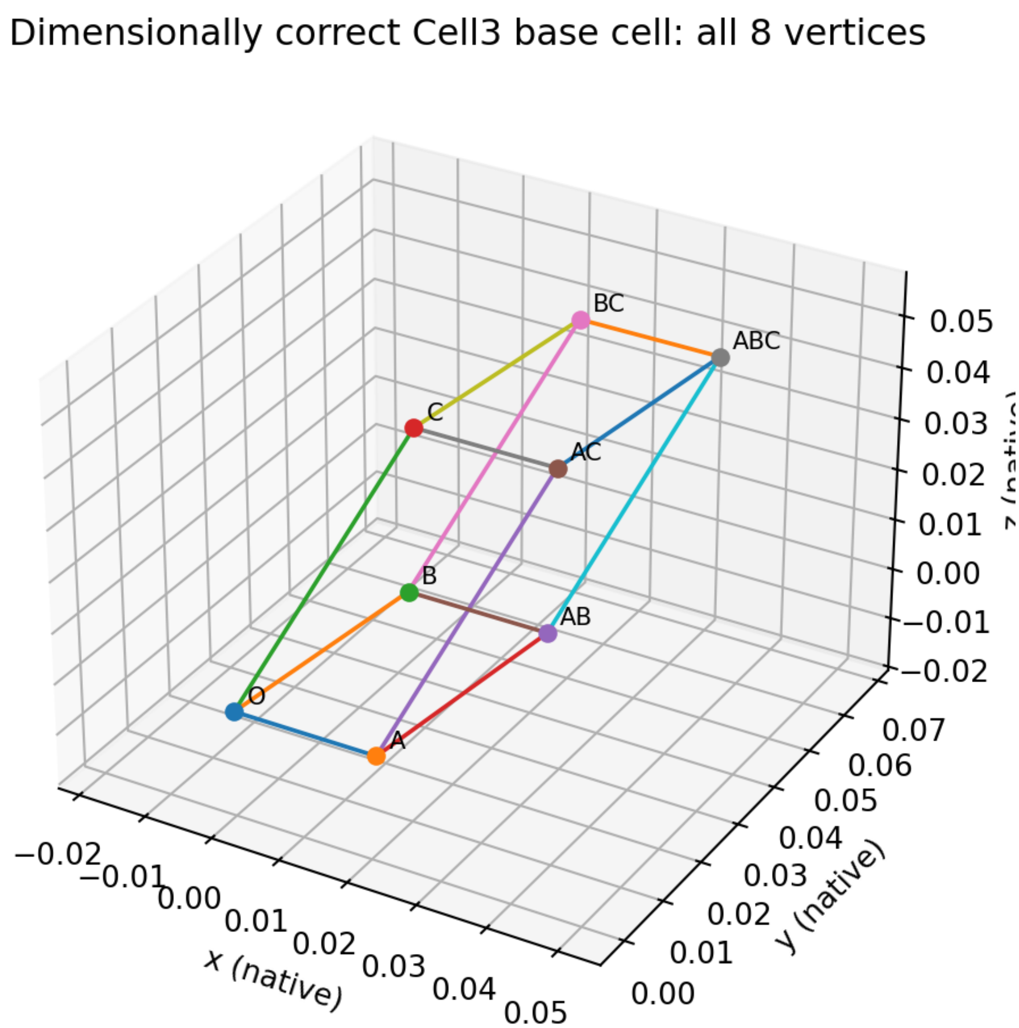

Figure 1. Dimensionally correct Cell3 base cell. All eight vertices are generated from the authoritative vectors and their sums.

Quantity	Value	Unit
|a|	0.0207970587	native length
|b|	0.0364942052	native length
|c|	0.049308566	native length
theta_ab	83.933483	deg
theta_ac	80.44813	deg
theta_bc	50.363232	deg
A_ab	0.00075472178809651	native length^2
A_ac	0.0010112557556791	native length^2
A_bc	0.0013857844409095	native length^2
Volume	2.8420669452112e-05	native length^3

2.1 Centered 27-node registry

The centered half-step registry uses addresses (i,j,k) in {-1,0,+1}^3. Every nonzero address has a unique opposite address. The registry therefore consists of one center and thirteen antipodal pairs.

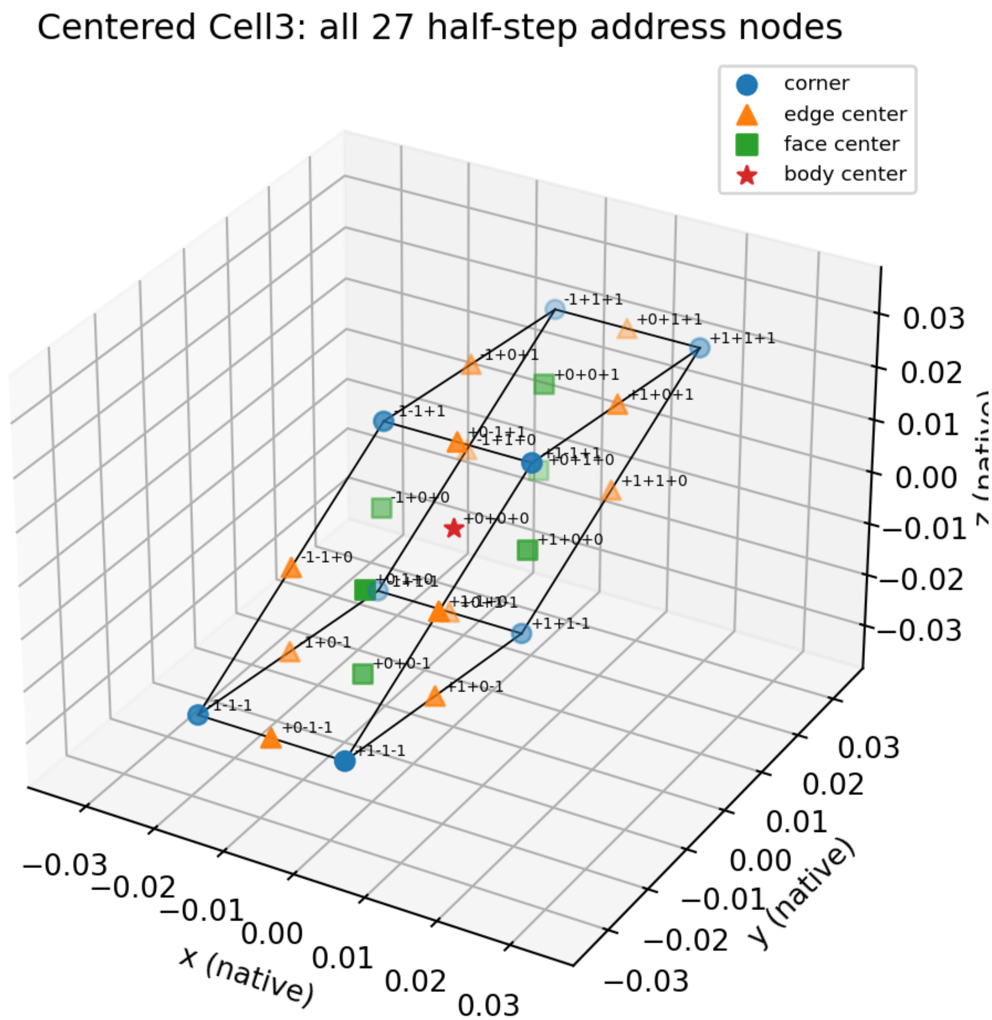

Figure 2. All 27 centered Cell3 nodes labeled by address.

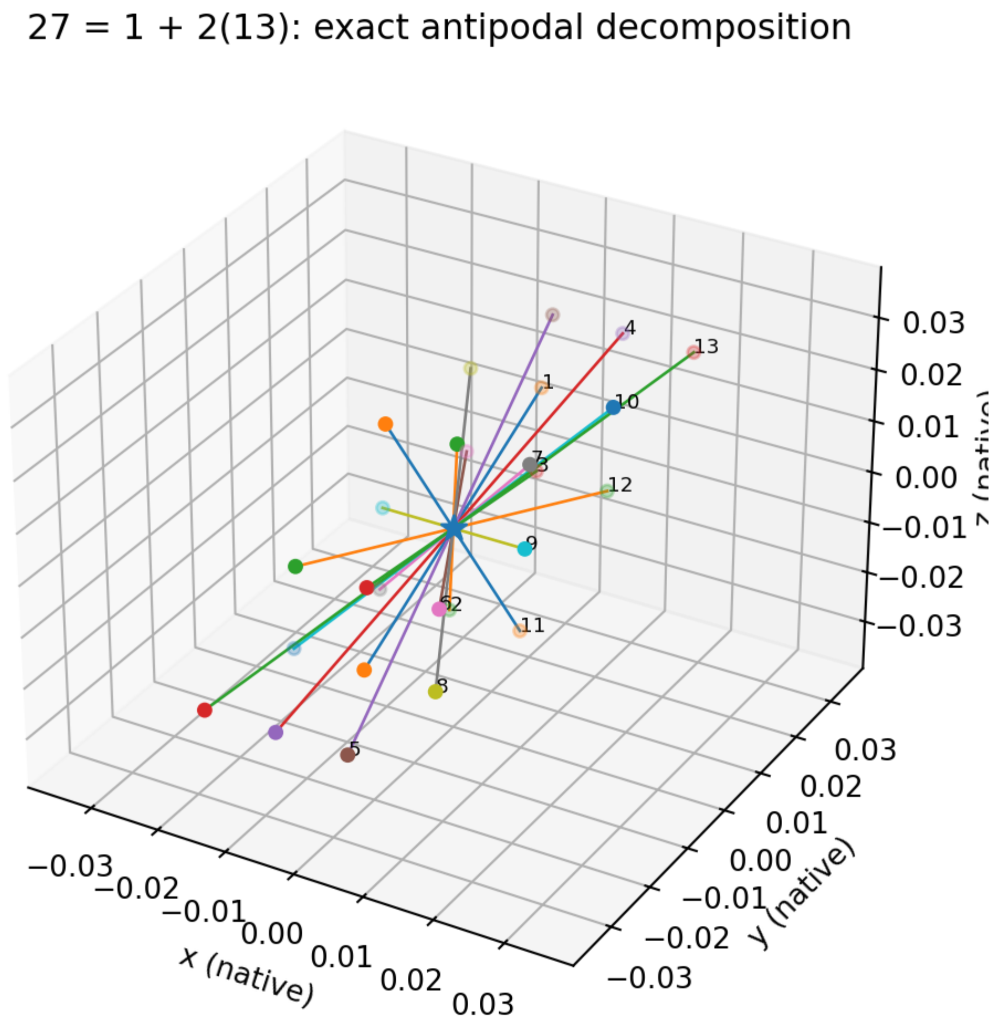

Figure 7. The exact decomposition 27=1+2(13).

2.2 Locked study planes

The theorem archive supplies a locked study-coordinate realization. The three central coordinate planes are shown together with all nodes. A rigid coordinate rotation preserves lengths, areas, volumes, radial shells, and antipodal relations.

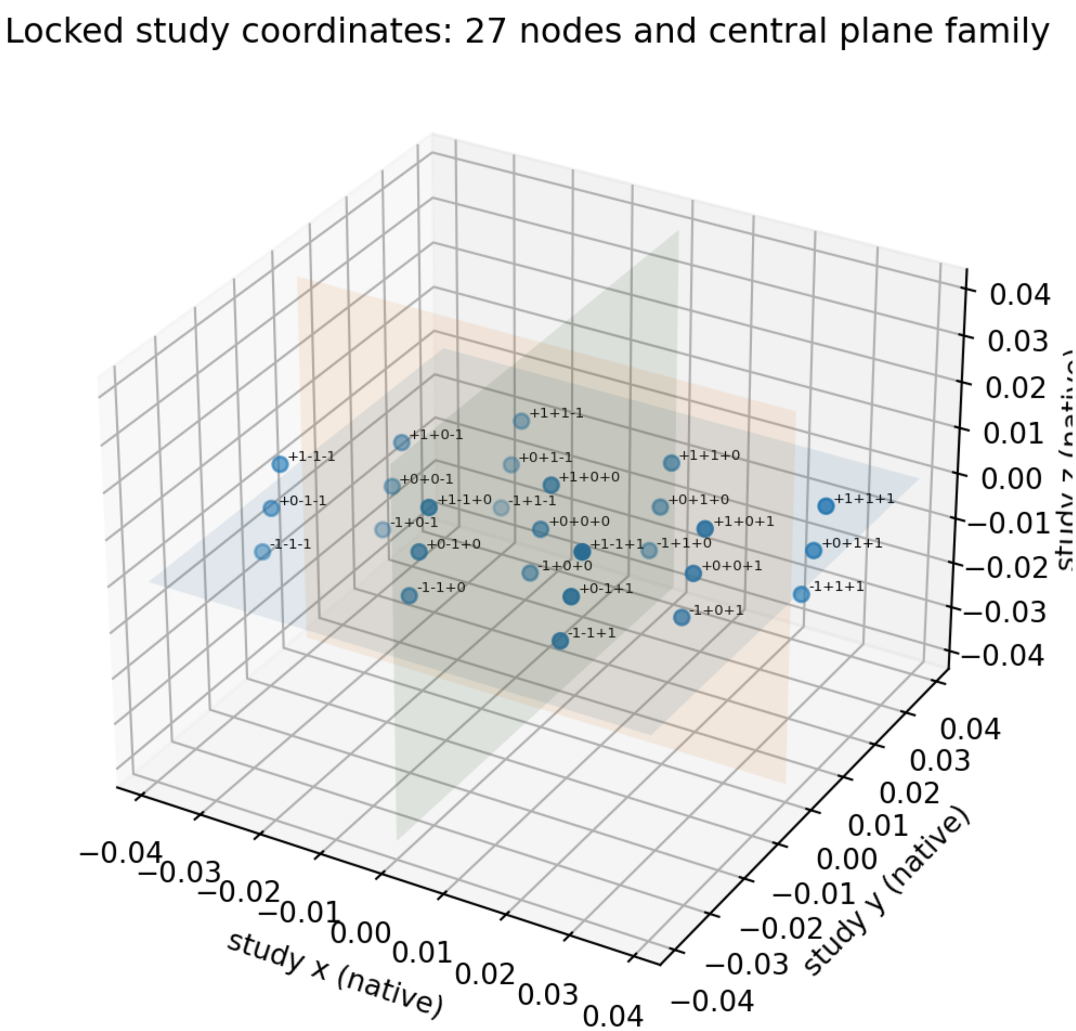

Figure 3. Locked study coordinates, all nodes, and the three central planes.

2.3 Dimensionally correct face planes

Each face figure uses one common local scale on both axes. Therefore edge lengths, included angles, and area are represented without aspect distortion.

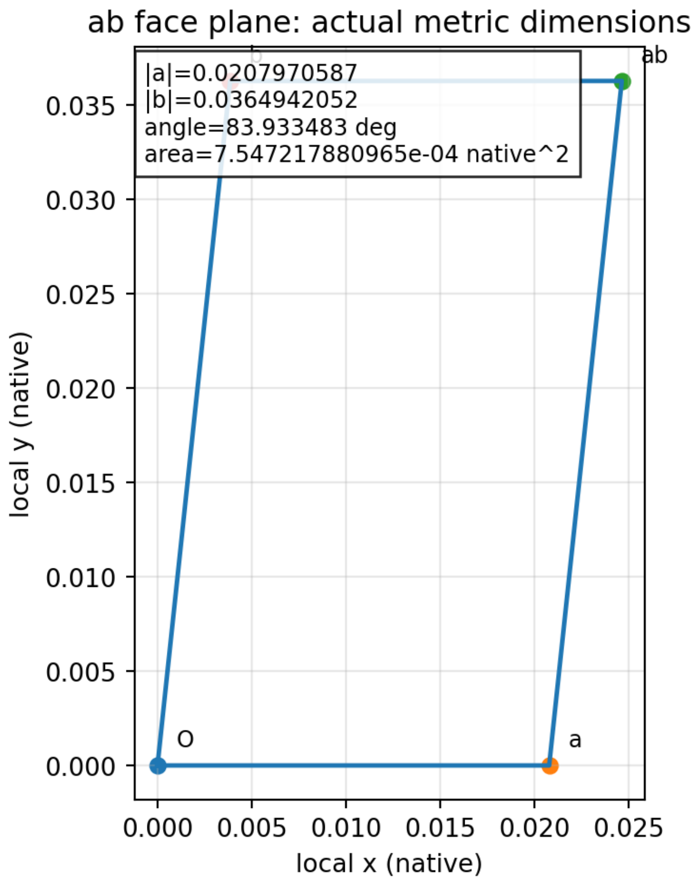

Figure 4. The ab face plane with all vertices and actual metric dimensions.

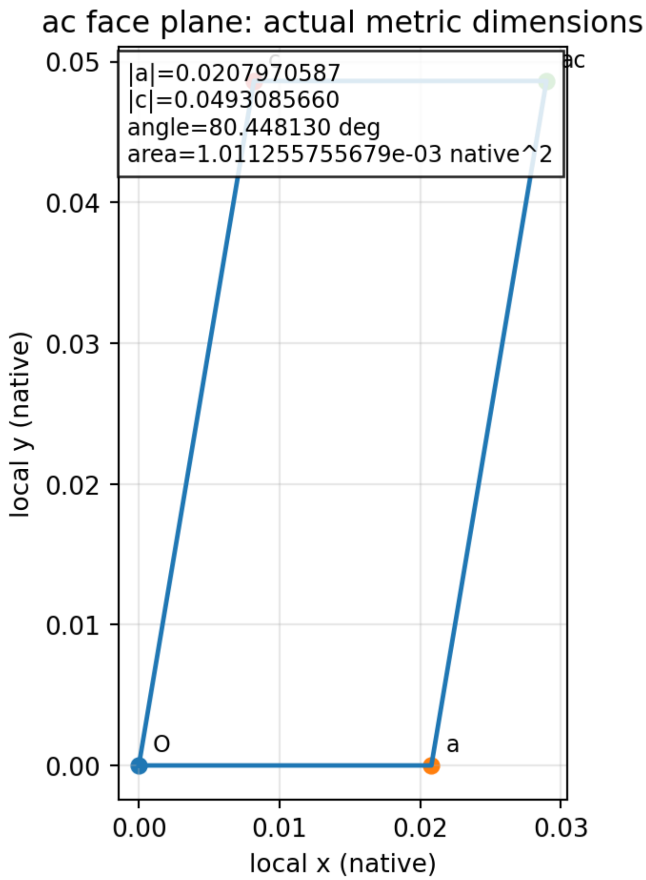

Figure 5. The ac face plane with all vertices and actual metric dimensions.

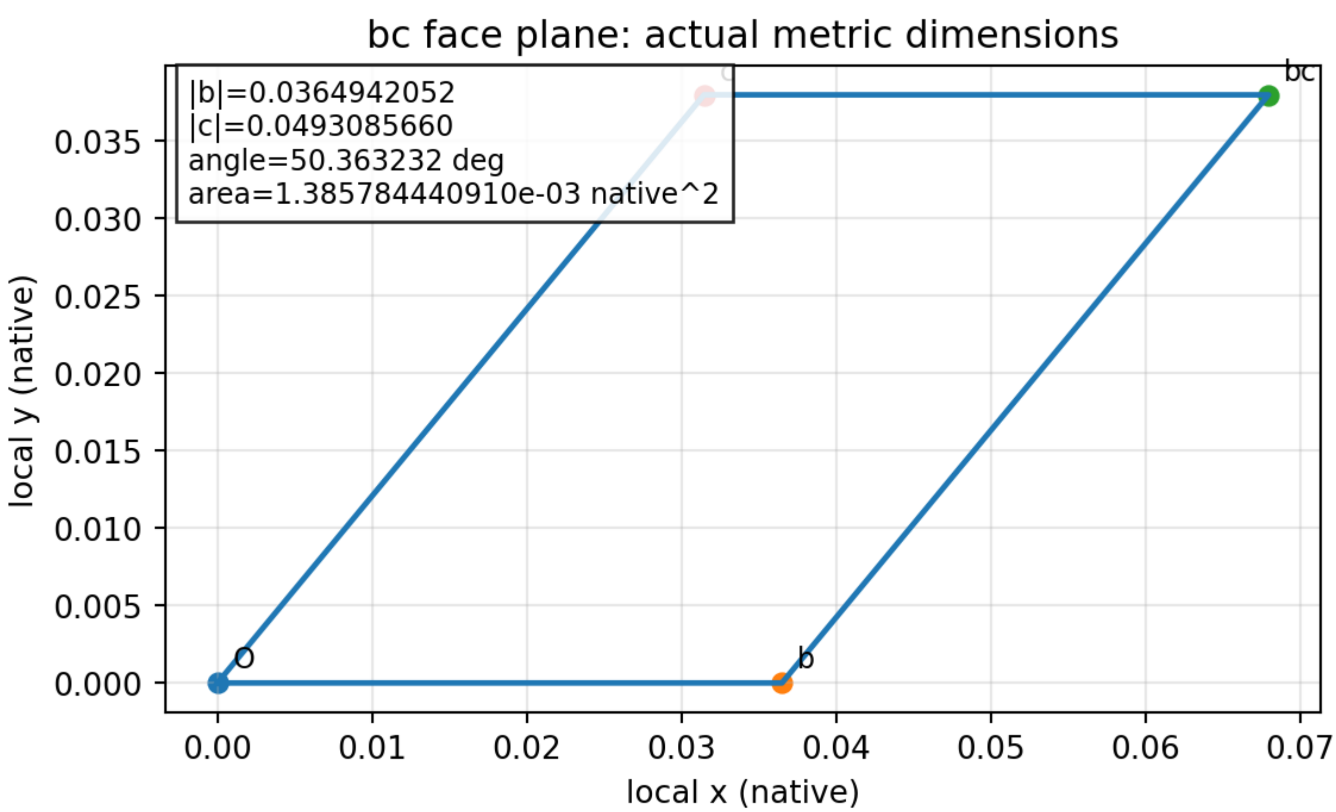

Figure 6. The bc face plane with all vertices and actual metric dimensions.

3. Antipodal radial geometry and parity

For a Cell3 lattice center $p(n)=B\,n$, the squared radius is $n^T G\,n$. The antipodal theorem states that, under radial injectivity, an occupied origin-centered sphere contains exactly p and $-p$. The finite-domain certificate in the source archive contains 3,381,312 occupied shells, each with multiplicity two, and no non-antipodal equal-radius shell.

$$p(-n)=-p(n)$$
$$\|p(n)\|^2=n^T G\,n$$
$$S_R \cap \Lambda=\{p,-p\}$$

The inversion operator exchanges the two members of each pair. Every pair supplies one even combination and one odd combination. The center is even. Therefore the finite 27-node representation has fourteen even dimensions and thirteen odd dimensions.

$$\dim(V_{even})=14$$
$$\dim(V_{odd})=13$$

Counterrotation belongs to the odd sector because reversing the pair changes the sign of angular momentum. Common translation belongs to an even sector because both masses move in the same laboratory direction. The forward force process is therefore a parity-conversion map from an odd internal mode to an even external mode.

4. Mechanical counter-rotating doublet

4.1 Inertia, energy, and angular momentum

An axisymmetric rotor is represented by $I=\beta\,m\,r^2$. Rotational energy is even under reversal, while angular momentum is odd.

$$I=\beta\,m\,r^2$$
$$E=(1/2)I\,\omega^2$$
$$L=I\,\omega$$

For a reaction-balanced pair, the angular momenta are equal in magnitude and opposite in sign.

$$L_i=+L$$
$$L_o=-L$$

The total mechanical angular momentum is zero, but the stored energy remains positive because both energy terms contain L squared.

4.2 Source K-Field rotor response

The supplied rotor ledger uses the vector force law

$$F_j=(\kappa_F\,m_j/2)(\Omega_j \times g)$$

The response is zero when the spin axis is parallel to g and maximal when it is perpendicular. For the counter-rotating pair, the two force contributions point oppositely. Their magnitudes cancel only under the m omega condition, not under equal energy and not under equal angular momentum.

$$F_{net}=C_K(m_i\,\omega_i-m_o\,|\omega_o|)$$

4.3 Forward doublet transformation

Using $\omega_i=L/I_i$ and $|\omega_o|=L/I_o$ gives

$$F_K=(\kappa_F\,L/2)[1/(\beta_i\,r_i^2)-1/(\beta_o\,r_o^2)](n \times g)$$

Define the reciprocal specific areal inertias a_i and a_o, the geometric contrast Gamma, and the active vector b.

$$a_i=1/(\beta_i r_i^2)$$
$$a_o=1/(\beta_o r_o^2)$$
$$\Gamma=(\kappa_F/2)(a_i-a_o)$$
$$b=n \times g$$

$$F_K=\Gamma L b$$

The residual vanishes only at the geometric double-null $\beta_i r_i^2=\beta_o r_o^2$, or at the orientation null n parallel to g.

5. Algebraic invertibility

For a fixed doublet orientation, the forward map has a scalar inverse whenever Gamma is nonzero and b is nonzero.

$$L=(F_K \cdot b)/(\Gamma |b|^2)$$

If the measured linear acceleration is $a=F_K/M$, then

$$L=M(a \cdot b)/(\Gamma |b|^2)$$

For one rotor, the transverse angular velocity can be recovered from the force vector by

$$\Omega_\perp=[2/(\kappa_F M |g|^2)](g \times F)$$

The component of Omega parallel to g remains undetermined because it produces no force. The spatial map therefore has one null dimension and two active transverse dimensions. Invertibility is exact only on this two-dimensional active subspace.

6. Why algebraic inversion is not yet a dynamical law

An algebraic inverse identifies which angular momentum corresponds to a measured force, but it does not determine how translation changes angular momentum with time. The source archives state rotation-to-force coupling. They do not state a translation-to-torque equation. A reciprocal torque law must therefore be supplied by an additional physical principle.

The minimal closure used here imposes five requirements: the same geometric coefficient appears in both directions; the law is local and linear in the active variables; reversal of travel, spin axis, or channel ordering reverses the induced torque; the forward source equation is reproduced exactly; and total translational-plus-rotational power is conserved in the absence of external work and dissipation.

Reciprocal-model status
The equations derived from these requirements are a disciplined reciprocal completion of the source law. They are not an additional equation quoted from the source archive. Their validity is experimentally testable by the motion protocols developed below.

7. Minimal local lossless reciprocal completion

7.1 Odd internal coordinate

Let q be an antipodal internal coordinate defined by $\dot{q}=L$. Because the two rotors carry +L and -L, their combined rotational kinetic energy is

$$T_{rot}=L^2/(2I_i)+L^2/(2I_o)=(1/2)A L^2$$

$$A=1/I_i+1/I_o$$

A is the positive rotational energy metric of the odd doublet mode.

7.2 Interaction Lagrangian

The minimal interaction term linear in translation and in the odd coordinate is

$$\mathcal{L}=(1/2)M|v|^2+(1/2)A \dot{q}^2-\Gamma q(v \cdot b)$$

This expression is dimensionally closed. Gamma has units of inverse area, q has units of angular momentum times time, and v dot b has units of velocity squared. Their product is energy.

7.3 Translational equation

The Euler-Lagrange equation for translation gives

```
M vdot=Gamma L b
```

which is exactly the established forward force law.

7.4 Reciprocal internal equation

The Euler-Lagrange equation for q gives

```
A Ldot=-Gamma(v dot b)
```

This is the reciprocal law. Translation projected onto the active direction generates equal-and-opposite angular momentum. Reversing travel, reversing n, or exchanging the inner and outer response weights reverses the sign of the induced doublet torque.

8. Exact power reciprocity and conserved energy

The translational power delivered by the rotating doublet is

```
P_trans=F_K dot v=Gamma L(b dot v)
```

The rate of change of rotational energy is

```
dT_rot/dt=A L Ldot=-Gamma L(b dot v)
```

```
P_trans+dT_rot/dt=0
```

The coupling therefore transfers energy reversibly between linear motion and antipodal rotation. It behaves as a lossless mechanical gyrator: one channel is an even translational variable, the other is an odd rotational variable, and the coupling changes sign under parity reversal while conserving total energy.

9. Acceleration, velocity, and displacement

Choose the unit active direction bhat=b/|b| and define u=v dot bhat and G=Gamma|b|. The coupled scalar equations are

```
M udot=G L
A Ldot=-G u
```

The reciprocal torque is directly proportional to directed velocity, not directly to acceleration. Acceleration is the externally controlled quantity that creates velocity and therefore appears when the rotational equation is differentiated.

```
Lddot+omega_R^2 L=-(G/A)a_ext
```

```
omega_R=|G|/sqrt(MA)
```

At early time under constant acceleration a0, u approximately equals a0 t and

```
L approximately -(G/(2A))a0 t^2
```

For a prescribed trajectory with negligible backreaction, integration gives

```
Delta L=-(G/A) integral u dt=-(G/A) Delta x_b
```

The accumulated doublet angular momentum is therefore proportional to directed displacement through the active field channel. Acceleration initiates the process, but traversal accumulates the rotation.

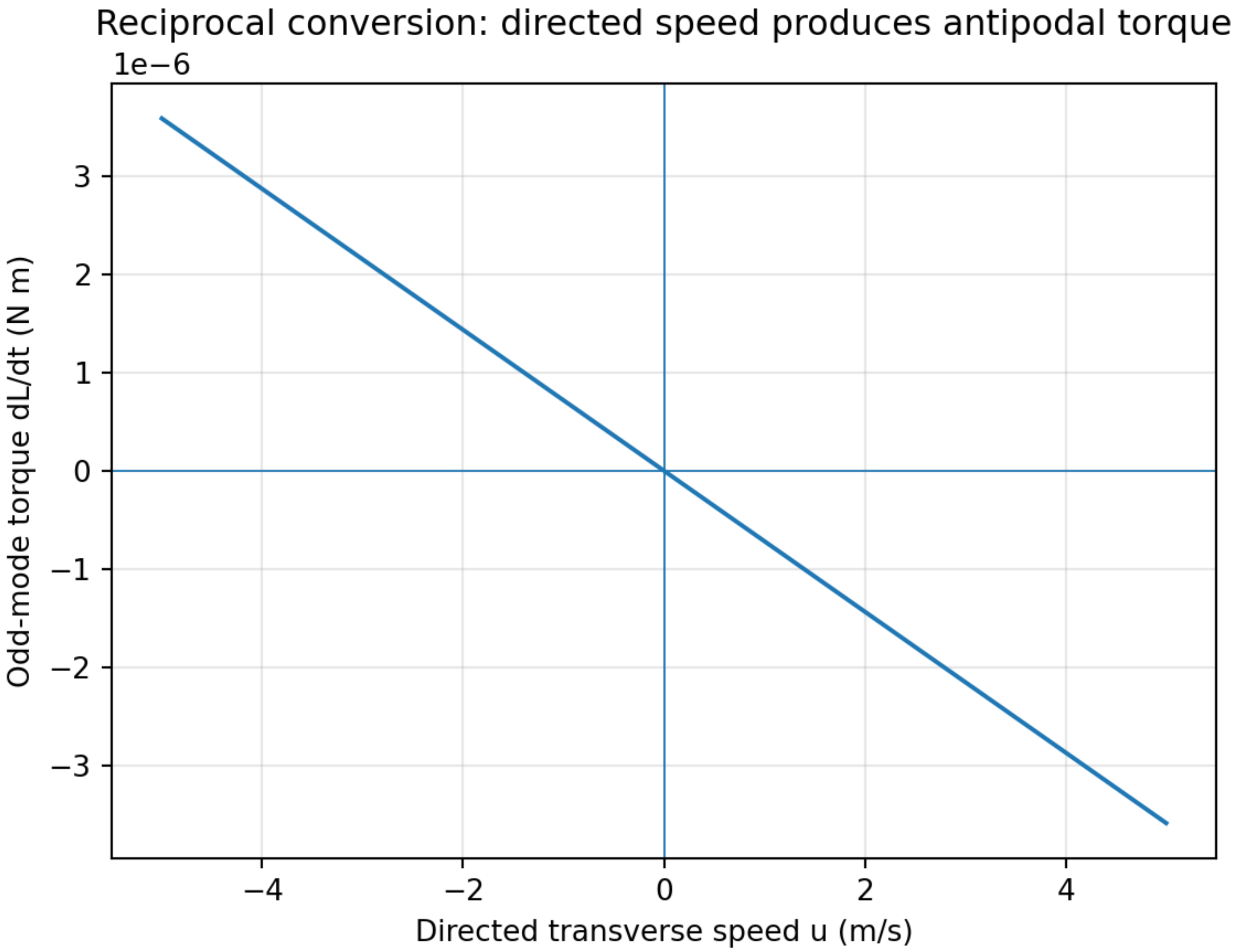

Figure 10. Reciprocal torque is linear in directed transverse speed.

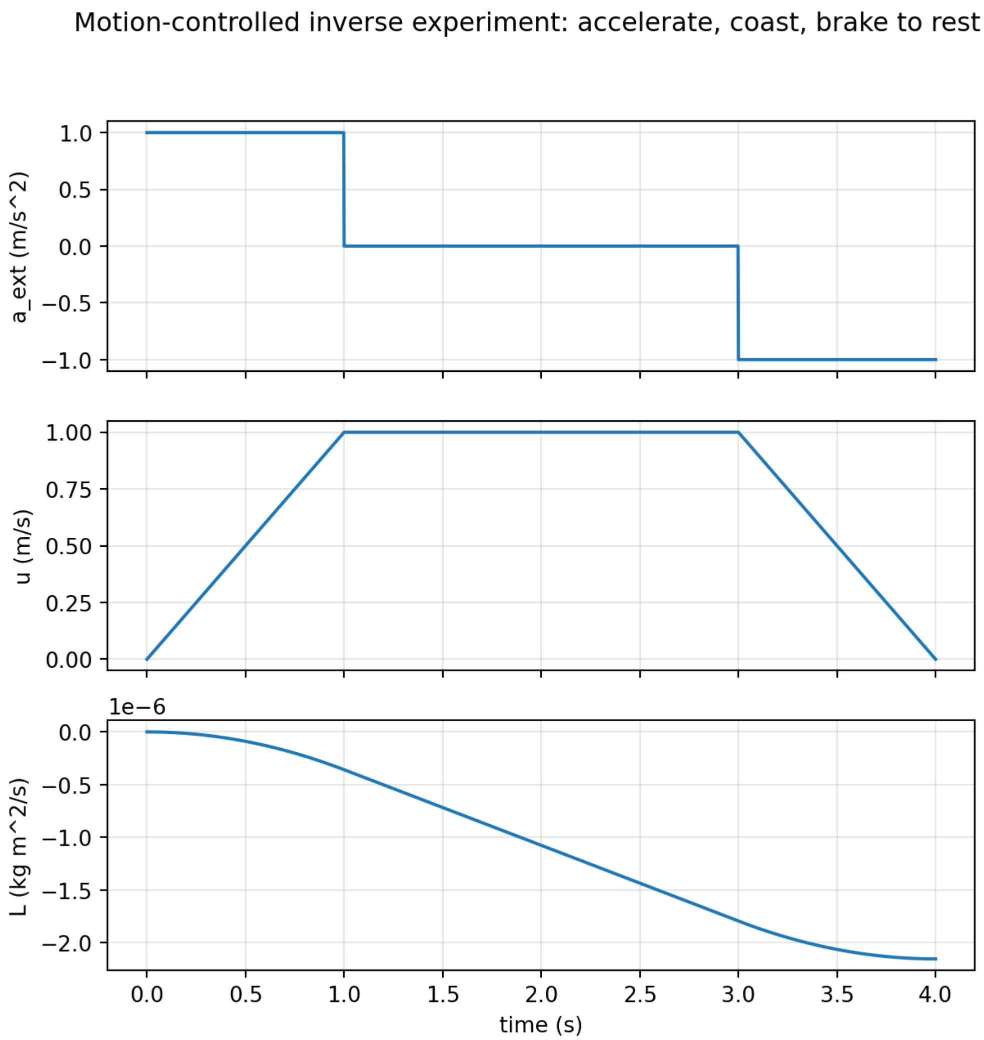

Figure 11. A computed accelerate-coast-brake trajectory. The final velocity is zero, but the nonzero displacement leaves a nonzero doublet angular momentum in the motion-controlled model.

10. Motion-controlled and force-controlled inversion

10.1 Motion-controlled trajectory

A translation stage prescribes u(t). The odd angular momentum follows directly from the path integral. If the stage accelerates, coasts, and brakes to rest at a displaced location, the final L remains nonzero. If the stage returns along the same path in a uniform fixed field, the signed displacement and final L return to zero.

10.2 Force-controlled response

With an external force F_ext, the translational equation becomes M udot=F_ext+G L. For a constant external force, the odd angular momentum obeys a forced oscillator equation. The static balance value is

```
L_eq=-F_ext/G=-M a_ext/G
```

At this state the generated K-Field force exactly opposes the applied force. A lossless system oscillates about this equilibrium. A dissipative implementation would approach it according to the separately specified damping law.

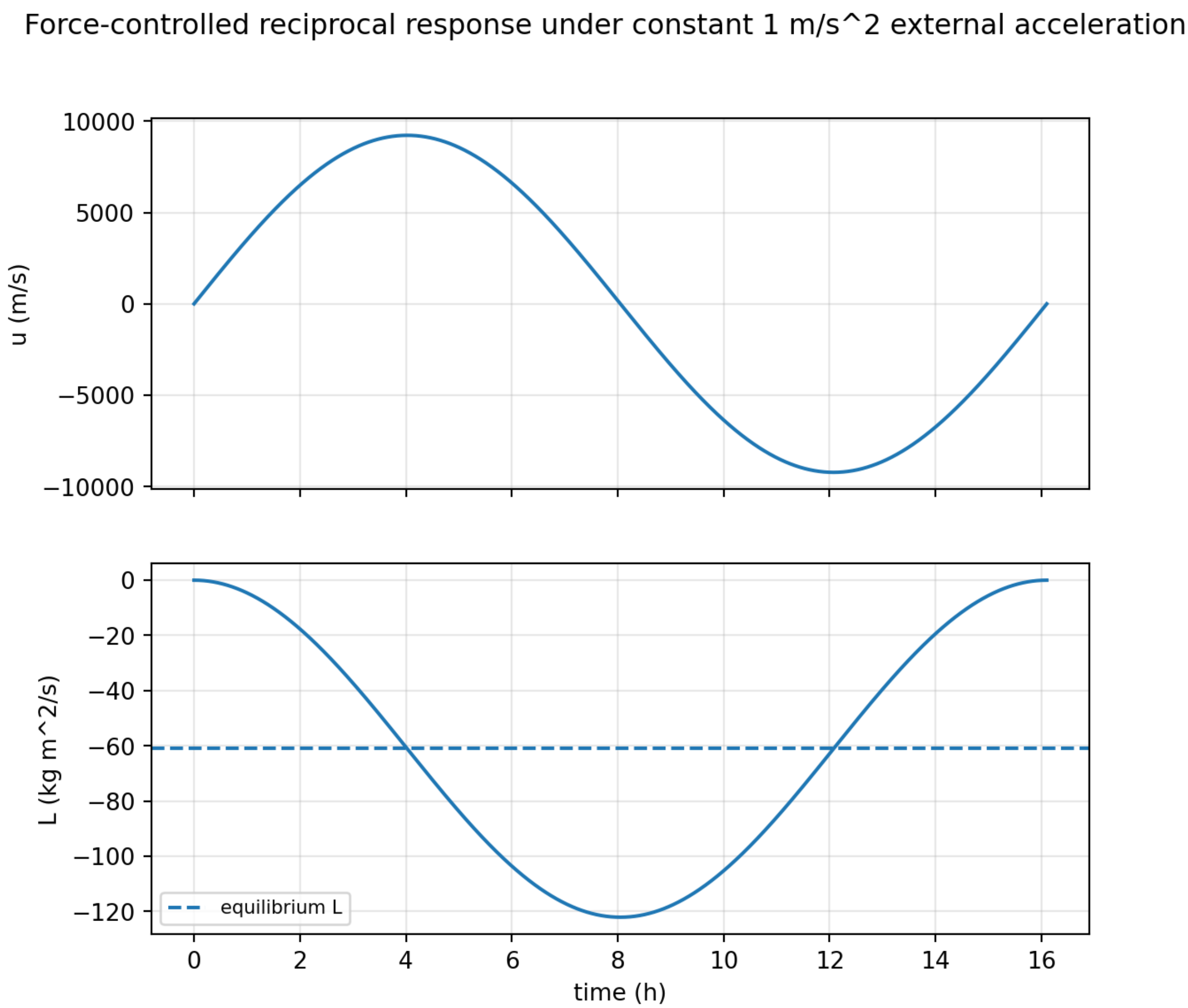

Figure 12. Force-controlled translation and doublet angular momentum over one lossless exchange period.

11. Free translation-rotation exchange eigenmode

With no external force, both active variables satisfy harmonic equations.

```
uddot+omega_R^2 u=0
Lddot+omega_R^2 L=0
```

The exact solutions are

```
u(t)=u0 cos(omega_R t)+[G L0/(M omega_R)]sin(omega_R t)
L(t)=L0 cos(omega_R t)-[G u0/(A omega_R)]sin(omega_R t)
```

A body beginning with translation but no rotation evolves toward counterrotation. A body beginning with counterrotation but no translation evolves toward linear motion. The energy exchange is exact and periodic in the ideal model.

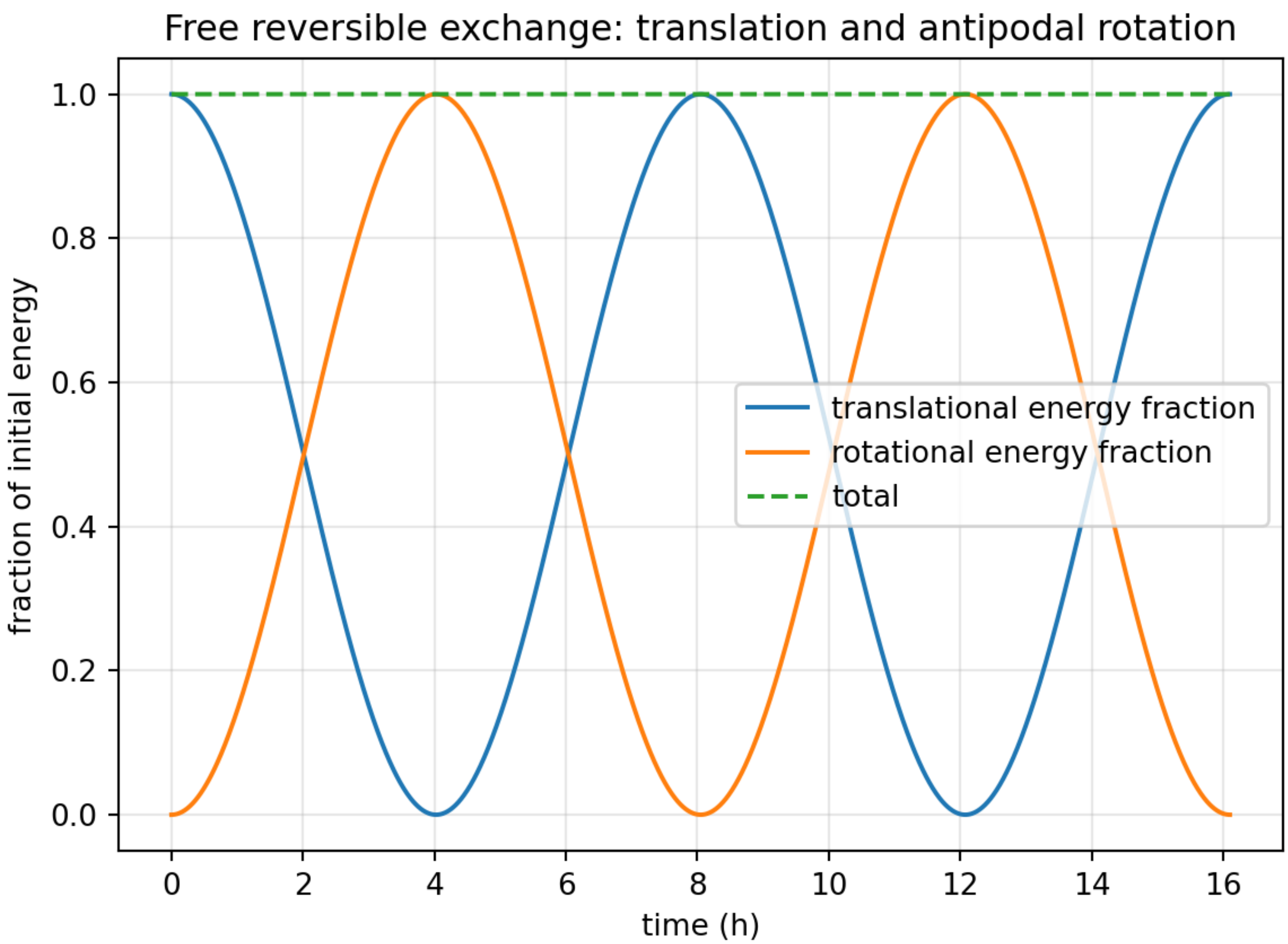

Figure 13. Translational and rotational energy exchange while total energy remains constant.

12. Worked numerical inversion using the source rotor

The source geometry uses a 0.1 kg inner thin ring of radius 0.02 m and a 1.0 kg outer thin ring of radius 0.10 m. Their moments of inertia are 4.0e-5 kg m^2 and 1.0e-2 kg m^2. The reaction-only source state carries L=0.4188790204786391 kg m^2/s in each opposite direction and produces a net force of 0.007545251040720908 N.

Quantity	Value	Unit
a_i	2500	m^-2
a_o	100	m^-2
scale ratio s	25	1
xi=(1/2)ln s	1.6094379124341	1
asymmetry tanh(xi)	0.92307692307692	1
A=1/I_i+1/I_o	25100	(kg m^2)^-1
G=F/L	0.01801296	(m s)^-1
|dL/dt|/|u|	7.1764780876494e-07	N m per (m/s)
omega_R	0.00010840564088548	s^-1
T_R	57959.94798663	s
T_R	16.0999855518	h

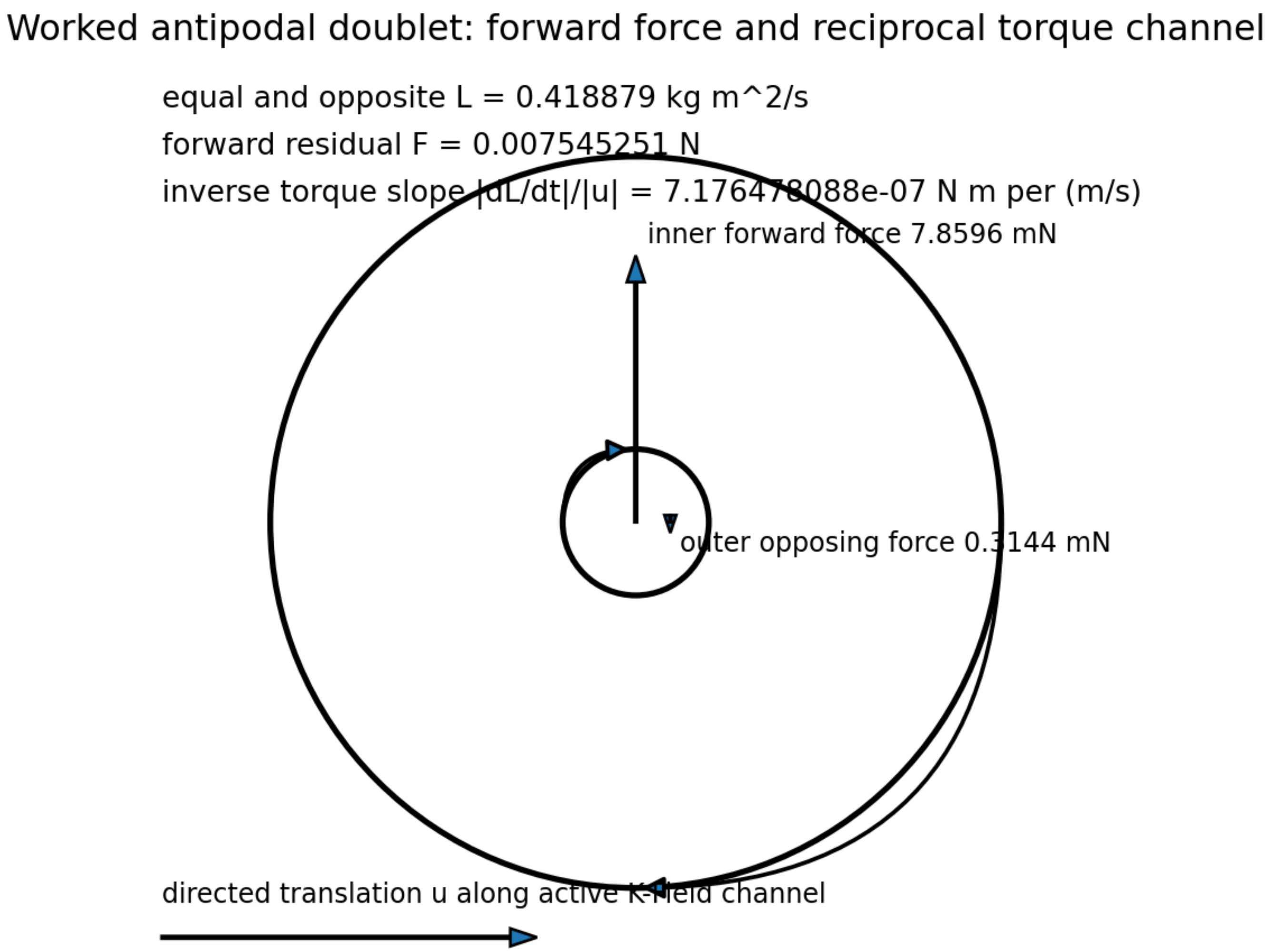

Figure 8. Worked source doublet with forward and reciprocal coefficients.

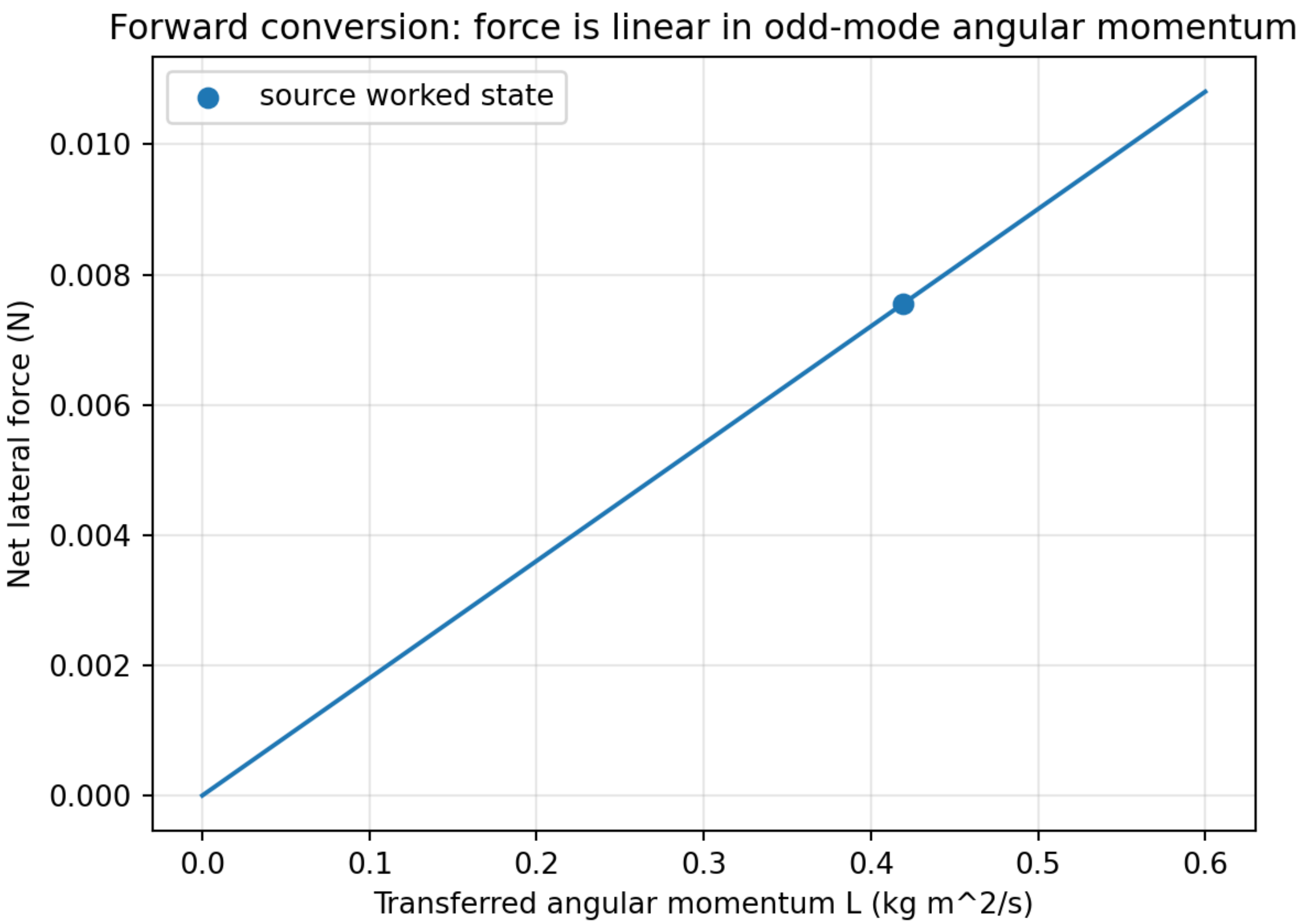

Figure 9. The source forward coefficient is the slope F/L.

12.1 One-second acceleration example

For a motion-controlled acceleration of 1 m/s^2 lasting one second from rest, the displacement is 0.5 m. The weak-backreaction prediction is

```
|L|=3.588239043825e-07 kg m^2/s
```

The corresponding inner angular speed is 8.970597609562e-03 rad/s, or 0.085662897 rpm. The outer angular speed is smaller by the inertia ratio and is opposite in sign.

12.2 Accelerate-coast-brake example

The plotted trajectory travels 3.000000 m and returns to zero velocity. Its final odd angular momentum is -2.152943426295e-06 kg m^2/s. The inner and outer angular velocities are -5.382358565737e-02 rad/s and 2.152943426295e-04 rad/s respectively. This retained rotation distinguishes displacement-coupled reciprocity from a torque law depending only on instantaneous acceleration.

13. Conditions for pure counterrotation

A general two-rotor assembly has an even co-rotating coordinate and an odd counter-rotating coordinate.

```
omega_+=(omega_i+omega_o)/2
omega_-=(omega_i-omega_o)/2
```

The reciprocal law derived above acts on the odd coordinate only because q was defined as the reaction-constrained antipodal mode. A freely mounted pair without such a constraint may also acquire an even component. Pure doublet generation therefore requires a differential gear, a reaction linkage, antipodally reversed local coordinates, a material mode that forbids the even channel, or active control maintaining L_i+L_o=0.

The experimental observables should separate total angular momentum and odd angular momentum:

```
L_total=L_i+L_o
L_odd=(L_i-L_o)/2
```

A true inverse antipodal response appears in L_odd while L_total remains at the mechanically imposed null.

14. Shared forward-inverse nulls and selection rules

Forward and inverse processes share the same geometric nulls. Coupling vanishes when the specific areal inertias are equal, when the spin axis is parallel to g, or when translation has no component along n cross g.

```
beta_i r_i^2=beta_o r_o^2 -> G=0
n parallel g -> b=0
v dot (n cross g)=0 -> Ldot=0
```

The same coefficient changes sign when inner and outer geometry are exchanged. Therefore swapping the channels or reversing the travel direction must reverse the induced spin.

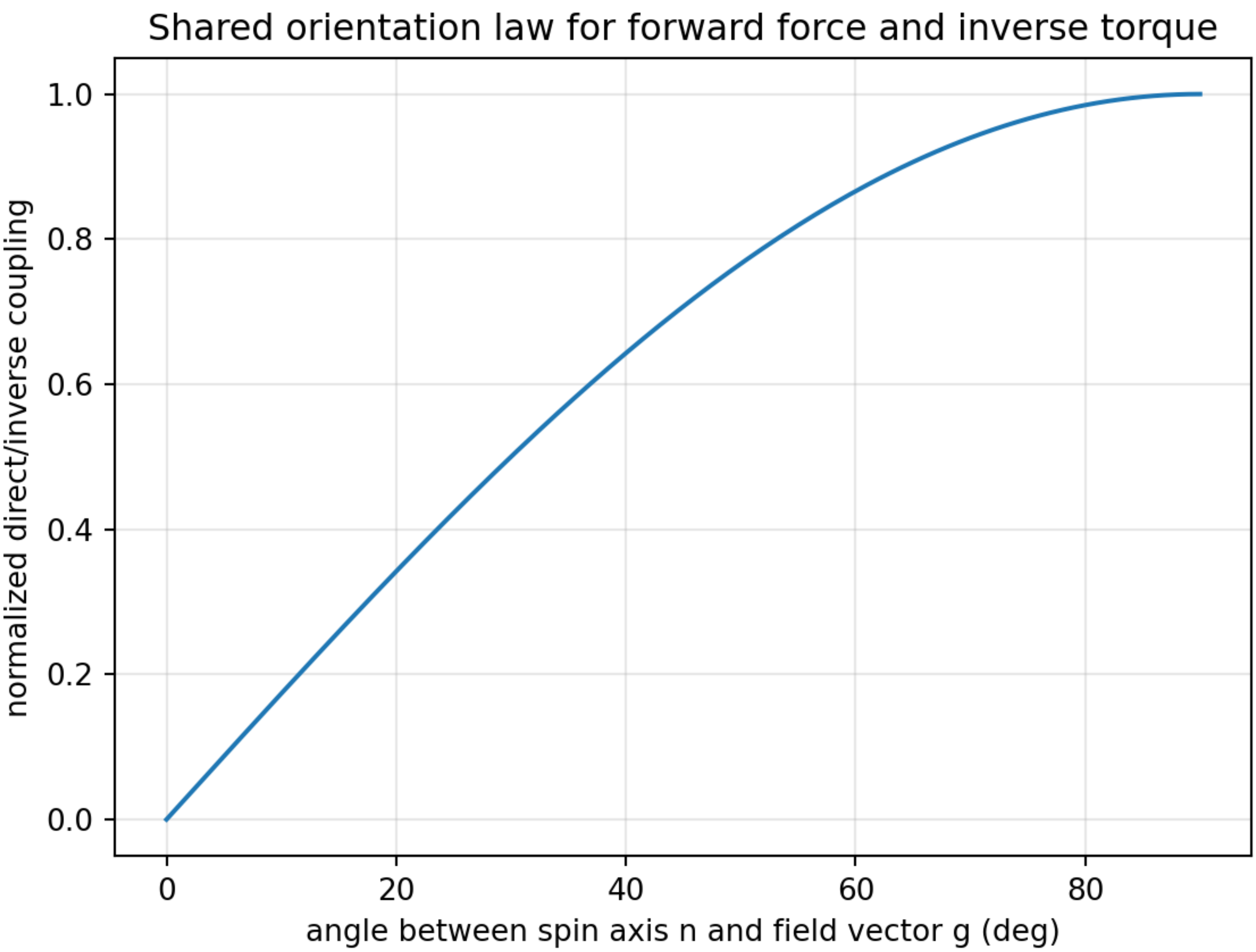

Figure 14. The direct and reciprocal coupling share the cross-product orientation factor sin(theta).

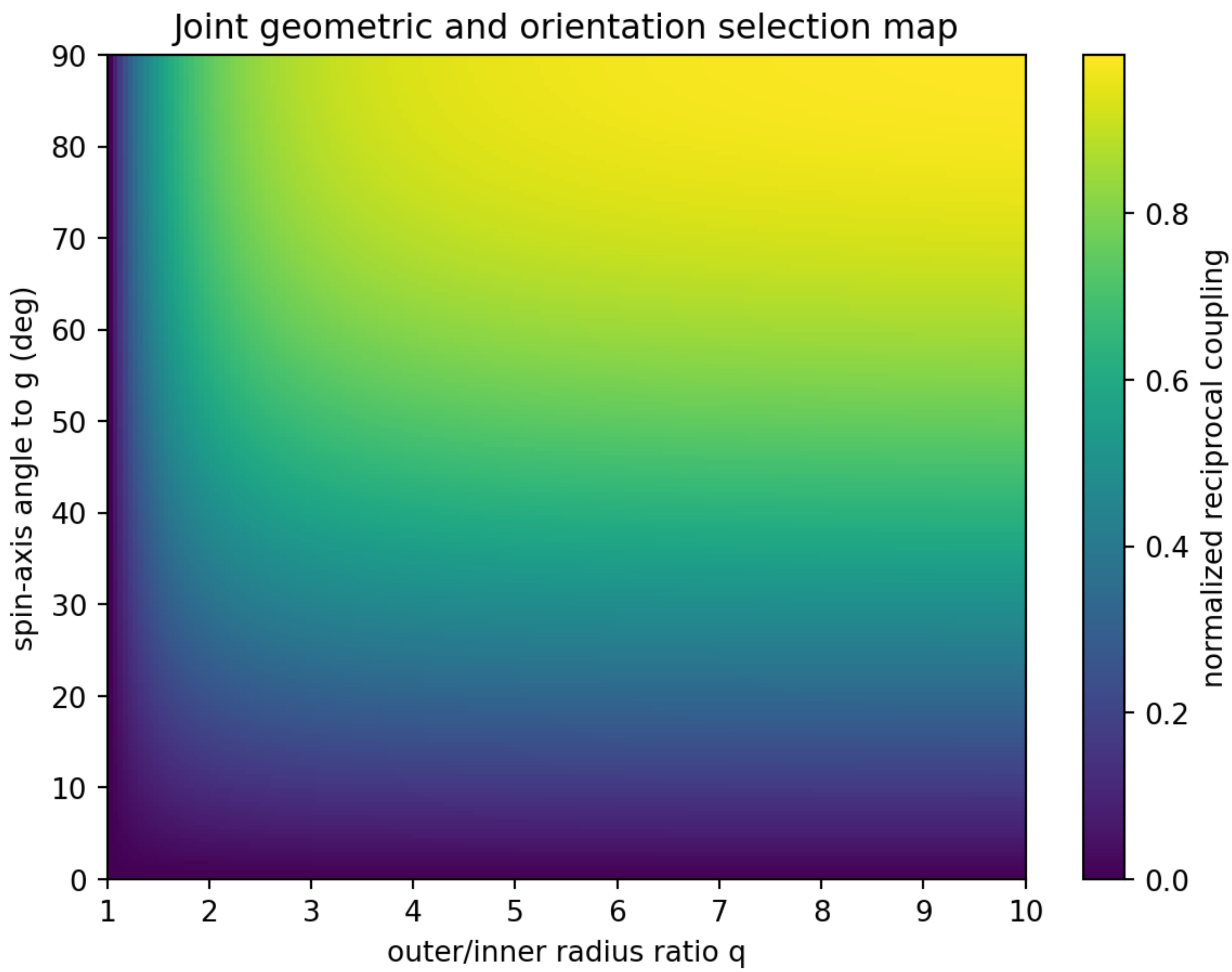

Figure 16. Joint radius-ratio and orientation map for identical rotor types.

15. Cell3 reciprocal extension

Let ell in R^13 contain the thirteen odd Cell3 amplitudes. Let A be their positive rotational-energy metric and K the map from odd internal amplitudes into physical force. The reciprocal completion is

```
M vdot=K ell
A elldot=-K^T v
```

The normal modes satisfy the generalized eigenproblem

```
K^T M^-1 K phi=omega^2 A phi
```

This is the higher-dimensional form of the two-variable exchange oscillator. It identifies which odd Cell3 combinations couple most strongly to each active translational direction and which combinations remain dark.

15.1 Bright and dark dimensions

At fixed g, the cross-product response has only two active spatial dimensions. Therefore rank(K) is at most two. Among the thirteen odd internal dimensions, at most two independent combinations can be bright at one fixed orientation and at least eleven are dark. The spatial count is two active transverse directions and one field-parallel null direction.

```
rank(K)<=2
bright odd dimensions<=2
```

```
dark odd dimensions>=11
```

Rotating the apparatus, changing the external field vector over time, or commutating among Cell3 channels changes K. A mode dark in one configuration can become bright in another.

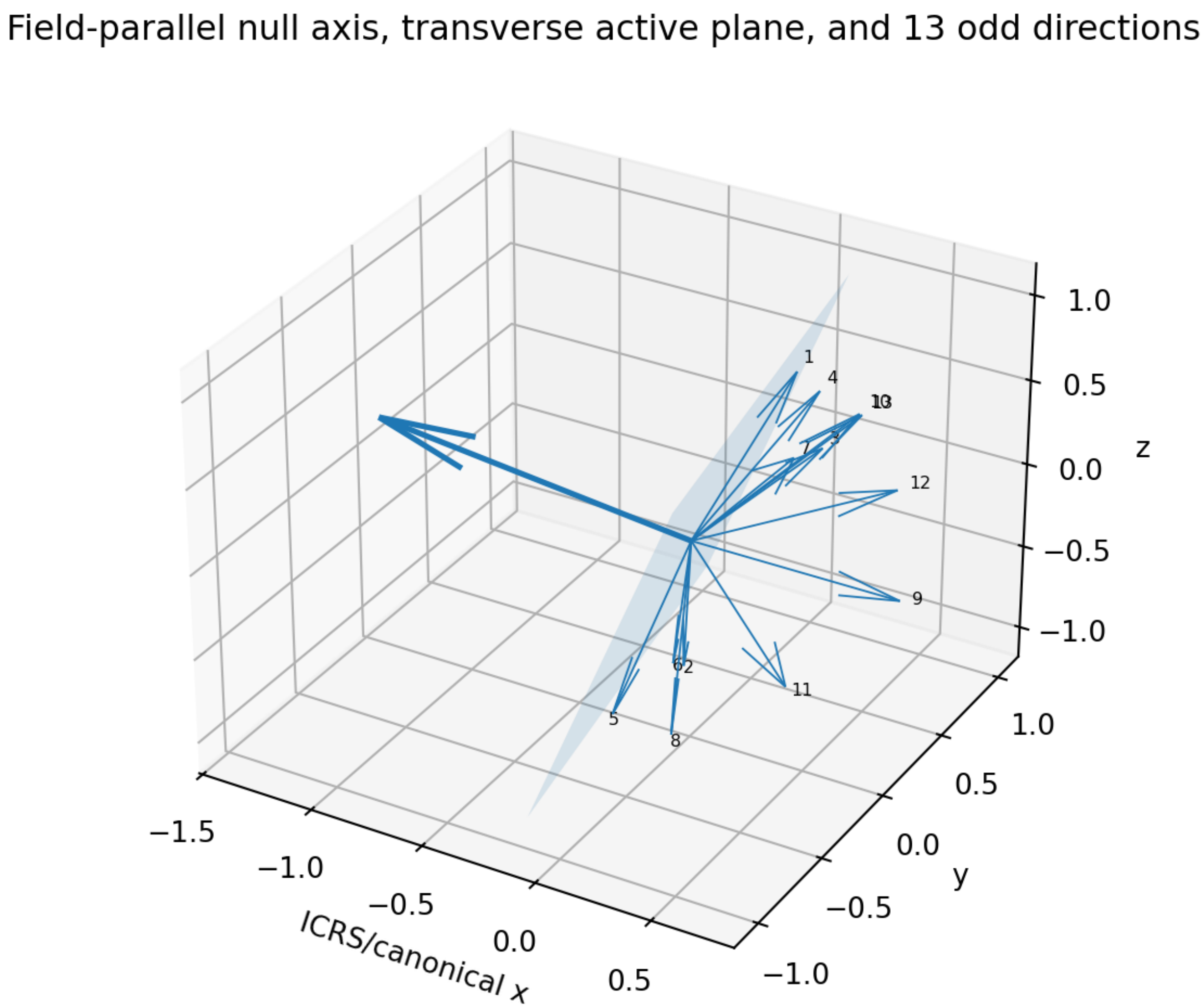

Figure 15. The source field direction, its transverse active plane, and the thirteen actual Cell3 antipodal directions.

16. Cell3 metric eigen-dimensions

Diagonalization of the source Gram metric gives three positive eigenvalues and three principal lengths. These are the intrinsic metric axes of the nonorthogonal cell.

Mode	Eigenvalue (native^2)	Principal length (native)	Eigenvector in (a,b,c) coefficients
1	0.00041706288436338	0.02042211752888	(0.9907217506, 0.063748854, -0.1200270655)
2	0.0006115114630215	0.02472875781396	(0.1175940076, -0.8448481837, 0.5219225957)
3	0.0031671049858395	0.056277037820406	(-0.06813268092, -0.5311945313, -0.8445059548)

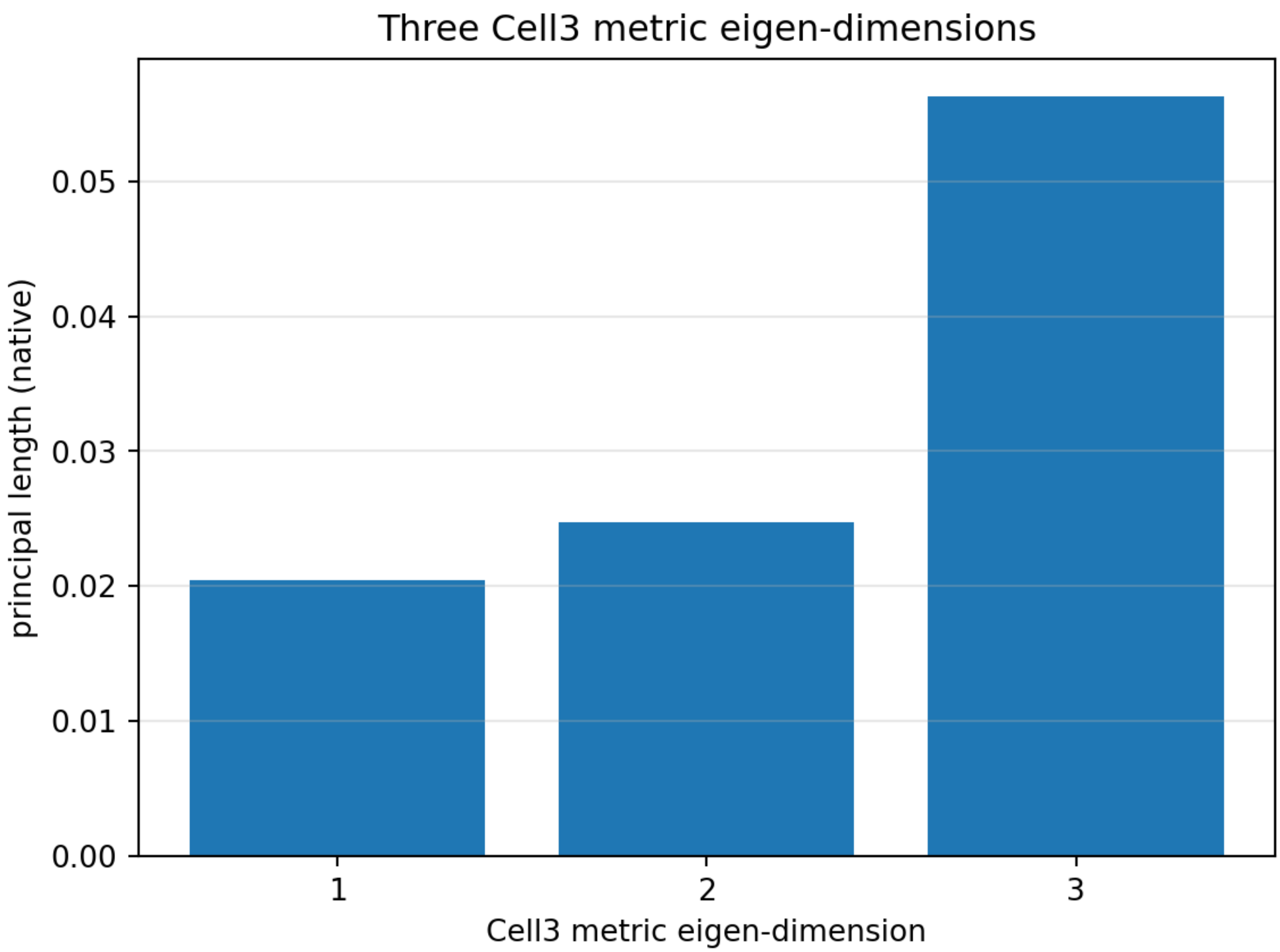

Figure 17. The three principal metric lengths of Cell3.

Combining thirteen odd inversion channels, three metric directions, and two transverse helicities provides a structured tensor-product space for reciprocal actuation. The exact physical selection rules must be determined by experiment and by the detailed construction of K.

17. Decisive experimental protocols

The inverse model is experimentally distinguishable from a torque law depending only on acceleration. The key is to separate acceleration history, velocity, and displacement.

Motion program	Velocity-reciprocal prediction	Pure acceleration-coupling prediction
Accelerate, then coast	Torque continues during coast because u remains nonzero.	Torque disappears when acceleration becomes zero.
Accelerate and brake to rest at displaced position	Nonzero final L proportional to displacement.	Zero net L when final velocity equals initial velocity, absent memory.
Same displacement, different acceleration profiles	Same weak-coupling final L.	Generally different final L.
Same velocity change, different travel distance	Different final L.	Same final L.
Complete out-and-back path	Zero final L in uniform fixed field.	Depends on acceleration history.
Equal beta r^2 channels	Exact null.	A true reciprocal law must share this null.
Swap inner and outer geometry	Spin sign reverses.	No required sign reversal without geometric reciprocity.
Reverse travel direction	Spin sign reverses.	Sign follows acceleration only during transients.
Lock rotors	Differential reaction torque appears during translation.	Only transient torque if acceleration-only.

Most decisive single test
Accelerate, coast, and brake to rest at a nonzero displacement while measuring the odd rotor coordinate. The reciprocal model predicts retained counterrotation proportional to displacement even though the final translational velocity and acceleration are zero.

18. Measurement architecture

A practical inverse experiment should use two independently instrumented rotors joined by a differential constraint. Each rotor requires an angular encoder or optical phase readout. The carriage requires calibrated position, velocity, and acceleration records. The apparatus orientation relative to the field vector must be logged. The primary derived channels are L_total, L_odd, and the predicted path integral -(G/A)Delta x_b.

Measured quantity	Purpose
Rotor angles and angular velocities	Resolve co-rotating and counter-rotating components.
Carriage position	Tests the displacement integral directly.
Carriage velocity	Tests the instantaneous reciprocal torque law.
Carriage acceleration	Separates acceleration transients from coast response.
Constraint reaction torque	Detects inverse torque when rotors are locked or heavily damped.
Apparatus orientation	Tests the sin(theta) field selection rule.
Temperature, vibration, motor current	Rejects bearing, thermal, magnetic, and inertial artifacts.

18.1 Null sequence

A disciplined sequence begins with the equal-specific-inertia double-null, followed by the field-parallel orientation null, transverse translation, travel reversal, channel swap, out-and-back closure, and finally the unlocked accumulation test. Each operation changes one parity or geometry variable while preserving the rest of the apparatus.

19. Foundational implications

The inverse completion changes the doublet from a one-way force source into a reversible transducer. Rotation and translation are not independent outputs; they are conjugate energy channels coupled by an antisymmetric geometric operator. The forward direction converts odd internal angular momentum into even external force. The inverse direction converts even directed traversal into odd internal torque.

```
odd rotation -> even force
even translation -> odd torque
```

The scale contrast is the conversion element. Antipodal symmetry supplies the sign structure. The field vector supplies the spatial orientation. Cell3 supplies the finite odd address space. Together they define a reversible geometry in which translation can be stored as equal-and-opposite angular momentum and later returned to translation.

19.1 New pathways to discovery

The reciprocal model suggests inertial energy storage without net assembly angular momentum, displacement-integrating K-Field sensors, low-frequency translation-rotation exchange resonators, parity-selective materials, commutated access to dark Cell3 modes, and multi-axis transducers formed by combining two active transverse channels. The generalized eigenproblem provides the design framework for maximizing reciprocal coupling subject to stress, speed, energy, and geometric constraints.

20. Consolidated conclusions

The process is invertible on the active transverse subspace. The exact forward relation F=G L has an algebraic inverse whenever the geometric contrast and transverse orientation are nonzero. The source law alone does not determine reverse time evolution, but a minimal local lossless reciprocal completion is obtained from a Lagrangian using the same coefficient.

The resulting inverse law is A Ldot=-G u. Directed velocity is the immediate conjugate drive; acceleration creates that velocity, and displacement accumulates the induced odd angular momentum. Under force control, translation and doublet rotation exchange energy at omega_R=|G|/sqrt(MA). Under motion control, Delta L=-(G/A)Delta x.

Pure counterrotation requires an odd-mode constraint. Without it, an unconstrained pair can acquire a co-rotating component. Cell3 extends the pair into a thirteen-dimensional odd internal space. At fixed field orientation only two spatial combinations can be bright, leaving at least eleven dark modes.

The strongest experimental signature is retained counterrotation after an accelerate-coast-brake trajectory that ends at rest but at nonzero displacement, combined with exact reversal under travel reversal, geometry swap, and field-orientation nulling.

Appendix A. Compact formula ledger

ID	Equation	Meaning
A1	I=beta m r^2	Axial inertia
A2	F_j=(kappa_F m_j/2)(Omega_j cross g)	Source rotor force
A3	F_K=Gamma L b	Forward doublet map
A4	L=(F dot b)/(Gamma |b|^2)	Algebraic inverse
A5	A=1/I_i+1/I_o	Odd-mode energy metric
A6	calL=(1/2)M|v|^2+(1/2)A qdot^2-Gamma q(v dot b)	Reciprocal Lagrangian
A7	M vdot=Gamma L b	Forward dynamical equation
A8	A Ldot=-Gamma(v dot b)	Inverse torque equation
A9	M udot=G L; A Ldot=-G u	Active scalar system
A10	omega_R=|G|/sqrt(MA)	Exchange frequency
A11	Delta L=-(G/A)Delta x_b	Motion-controlled accumulation
A12	K^T M^-1 K phi=omega^2 A phi	Cell3 generalized eigenproblem

Appendix B. Complete 27-node coordinate ledger

Address	Kind	i	j	k	x native	y native	z native	x nm	y nm	z nm
-1,-1,-1	corner	-1	-1	-1	-0.0164180796	-0.03352610771	-0.01882857359	-0.2638156015	-0.5387177117	-0.3025488722
-1,-1,+0	edge center	-1	-1	0	-0.01232693741	-0.01814491652	0	-0.1980766622	-0.2915634583	0
-1,-1,+1	corner	-1	-1	1	-0.008235795218	-0.002763725333	0.01882857359	-0.1323377229	-0.04440920491	0.3025488722
-1,+0,-1	edge center	-1	0	-1	-0.01448967154	-0.01538119119	-0.01882857359	-0.2328287782	-0.2471542534	-0.3025488722
-1,+0,+0	face center	-1	0	0	-0.01039852935	0	0	-0.1670898389	0	0
-1,+0,+1	edge center	-1	0	1	-0.006307387158	0.01538119119	0.01882857359	-0.1013508996	0.2471542534	0.3025488722
-1,+1,-1	corner	-1	1	-1	-0.01256126348	0.002763725333	-0.01882857359	-0.2018419548	0.04440920491	-0.3025488722
-1,+1,+0	edge center	-1	1	0	-0.00847012129	0.01814491652	0	-0.1361030155	0.2915634583	0
-1,+1,+1	corner	-1	1	1	-0.004378979098	0.03352610771	0.01882857359	-0.07036407624	0.5387177117	0.3025488722
+0,-1,-1	edge center	0	-1	-1	-0.006019550252	-0.03352610771	-0.01882857359	-0.09672576264	-0.5387177117	-0.3025488722
+0,-1,+0	face center	0	-1	0	-0.00192840806	-0.01814491652	0	-0.03098682335	-0.2915634583	0
+0,-1,+1	edge center	0	-1	1	0.002162734132	-0.002763725333	0.01882857359	0.03475211595	-0.04440920491	0.3025488722
+0,+0,-1	face center	0	0	-1	-0.004091142192	-0.01538119119	-0.01882857359	-0.06573893929	-0.2471542534	-0.3025488722
+0,+0,+0	body center	0	0	0	0	0	0	0	0	0
+0,+0,+1	face center	0	0	1	0.004091142192	0.01538119119	0.01882857359	0.06573893929	0.2471542534	0.3025488722
+0,+1,-1	edge center	0	1	-1	-0.002162734132	0.002763725333	-0.01882857359	-0.03475211595	0.04440920491	-0.3025488722
+0,+1,+0	face center	0	1	0	0.00192840806	0.01814491652	0	0.03098682335	0.2915634583	0
+0,+1,+1	edge center	0	1	1	0.006019550252	0.03352610771	0.01882857359	0.09672576264	0.5387177117	0.3025488722
+1,-1,-1	corner	1	-1	-1	0.004378979098	-0.03352610771	-0.01882857359	0.07036407624	-0.5387177117	-0.3025488722
+1,-1,+0	edge center	1	-1	0	0.00847012129	-0.01814491652	0	0.1361030155	-0.2915634583	0
+1,-1,+1	corner	1	-1	1	0.01256126348	-0.002763725333	0.01882857359	0.2018419548	-0.04440920491	0.3025488722
+1,+0,-1	edge center	1	0	-1	0.006307387158	-0.01538119119	-0.01882857359	0.1013508996	-0.2471542534	-0.3025488722
+1,+0,+0	face center	1	0	0	0.01039852935	0	0	0.1670898389	0	0
+1,+0,+1	edge center	1	0	1	0.01448967154	0.01538119119	0.01882857359	0.2328287782	0.2471542534	0.3025488722
+1,+1,-1	corner	1	1	-1	0.008235795218	0.002763725333	-0.01882857359	0.1323377229	0.04440920491	-0.3025488722
+1,+1,+0	edge center	1	1	0	0.01232693741	0.01814491652	0	0.1980766622	0.2915634583	0
+1,+1,+1	corner	1	1	1	0.0164180796	0.03352610771	0.01882857359	0.2638156015	0.5387177117	0.3025488722

Appendix C. Numerical reciprocal ledger

Quantity	Value	Unit/definition
m_i	0.1	kg
r_i	0.02	m
I_i	4.00000000000000e-05	kg m^2
m_o	1	kg
r_o	0.1	m
I_o	0.01	kg m^2
C_K	7.50540000000000e-06	N s/kg
kappa_F	1.61263357733437e-11	1
|g|	9.30825217270519e+05	m/s
a_i-a_o	2400	m^-2
G	0.01801296	(m s)^-1
A	25100	(kg m^2)^-1
G/A	7.17647808764940e-07	N m per (m/s)
M	1.1	kg
omega_R	0.000108405640885479	s^-1
T_R	57959.9479866295	s

Appendix D. Reproducibility record

Role	Filename	SHA-256	Bytes
concentric_rotor	K-Field_concentric_rotor_lateral_force_analysis_1785171858(1).json	854bf593256b5ee18ca02a65eb1f202a449266b815314b39fbacbf7588e28673	765870
mass_bridge	K-Field_Compton_classical_radius_Mass_Bridge_triangle_1785175483(1).json	9301e9108b98c848a14a188b0858a81a7509b53e02cd5de4a255fe8a0a0ece93	42979
antipodal_theorem	K-Native_Cell3_antipodal_sphere_theorem1784492216(3).json	2c6331de64d582f58b287f8bda04ca2cf90e70e7eae1c1d463b5f1ec8cfa5005	14664
prior_antipodal_doublet	K-Field_counter-rotating_system_is_an_antipodal_doublet_1785179808.json	c6daf91e61ea826c7d8a610a7a629624dc40915d772c9eea11e6ed44c45fbb93	138920

The companion file K-Field_counter-rotating_system_inversion_1785182637.json contains all source inputs used in the report, the complete 27-node registry, the reciprocal model parameters, full time-series arrays for the dynamic figures, the formula ledger, source hashes, and the figure manifest.

Appendix E. Figure manifest

ID	Filename	Title	Data basis
1	figure_01_base_cell_8_vertices.png	Dimensionally correct Cell3 base cell with all eight vertices.	Mass-Bridge canonical vectors and vertex sums
2	figure_02_all_27_nodes.png	All 27 centered Cell3 nodes, labeled by address.	Mass-Bridge centered_half_step_nodes
3	figure_03_study_planes_nodes.png	All 27 theorem nodes in locked study coordinates with three origin-centered planes.	Native Cell3 theorem lucas_nodes
4	figure_04_face_ab.png	Dimensionally correct ab face with vertices, lengths, angle, and area.	Canonical vectors a and b
5	figure_05_face_ac.png	Dimensionally correct ac face with vertices, lengths, angle, and area.	Canonical vectors a and c
6	figure_06_face_bc.png	Dimensionally correct bc face with vertices, lengths, angle, and area.	Canonical vectors b and c
7	figure_07_13_antipodal_pairs.png	The centered Cell3 registry as 13 antipodal lines plus one center.	Centered node coordinates paired under p -> -p
8	figure_08_worked_forward_inverse_doublet.png	Worked source geometry annotated with forward force and reciprocal torque coefficients.	Rotor worked input plus reciprocal completion
9	figure_09_forward_force_vs_L.png	Forward force versus transferred angular momentum for the worked geometry.	F=G L with source coefficient
10	figure_10_inverse_torque_vs_speed.png	Reciprocal odd-mode torque versus directed transverse speed.	dL/dt=-(G/A)u using worked geometry
11	figure_11_accel_coast_brake_response.png	Computed acceleration, velocity, and accumulated odd angular momentum for a 3 m traversal.	Motion-controlled reciprocal equation with worked coefficient
12	figure_12_force_controlled_exchange.png	Lossless force-controlled response over one reciprocal exchange period.	Coupled ODE solution using M=1.1 kg and worked rotor geometry
13	figure_13_energy_exchange.png	Free exchange of energy between translation and doublet rotation.	Lossless reciprocal model with u0=1 m/s
14	figure_14_orientation_dependence.png	Normalized coupling versus spin-axis angle to the field vector.	Cross-product magnitude sin(theta)
15	figure_15_transverse_plane_13_modes.png	Field null axis and active transverse plane with the 13 antipodal Cell3 directions.	Source g vector and actual Cell3 pair directions
16	figure_16_scale_angle_coupling_map.png	Normalized coupling across radius ratio and field orientation.	(1-1/q^2) sin(theta) for identical rotor types
17	figure_17_metric_eigendimensions.png	Principal lengths obtained from the Cell3 Gram metric eigenvalues.	Theorem Gram matrix